FPT Software

Performance Test



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Fundamentals



What is Performance Testing?

Ensuring the software is fast, stable, and scalable under real-world workloads.

	Load Test	Stress Test	Spike Test	Soak Test
Goal	Evaluate performance under expected peak load.	Find the maximum capacity (breaking point) and observe recovery.	Test the ability to survive sudden, extreme bursts of traffic.	Evaluate stability over an extended period to find memory leaks
Question	Can we handle 100 normal users?	At what number of users will the system crash?	Can the system handle a sudden traffic burst?	Will the RAM get full if we run this for 3 hours?

Test Preparation

Provided Inputs	QA Definitions
1. API & Target Scope • API Spec: URL, Method, Headers, Request Body.	**1. Workload Model & Scenarios** • Concurrency: VUs, Ramp-up, Hold, Ramp-down time. • Test type: Load test (normal) or Stress test (peak).
2. Requirements (NFRs) • Expected RPS (>= 150 RPS). • Target Response Time (P95 <= 500ms).	**2. Pass/Fail Thresholds** • Success rate: Error rate <= 1%. • Performance target: P95 <= 500ms. • Auto-stop rule: Abort if error rate > 10%.
	3. Automation Script Scripting: K6 / JMeter execution scripts.

Structure of a Performance Test Plan

Structure of a Test Plan	Key Description
1. Introduction & Objectives	Define the test purpose
2. Scope (In / Out of Scope)	Specify what endpoints to test and what to exclude
3. Entry & Exit Criteria	Set conditions to start (scripts ready) and conditions to finish the test.
4. System Under Test (SUT)	Detail the environment and hardware limits
5. Workload Model & Scenarios	Configure user loads (VUs, duration) and test types (Load, Stress, Soak).
6. Pass/Fail Criteria	Set clear metric thresholds (P95 <= 500ms, Error Rate <= 1%).

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Metrics



Report Metrics

Metric Category	Actual Metric Name	What it Means
1. Traffic Volume	Total Requests (http_reqs)	Total number of requests sent during the test run.
	Throughput / RPS (http_reqs/s)	Average requests processed per second.
2. Response Time (Latency)	Average (avg)	The mean response time of all requests.
	Min / Max (min / max)	The fastest and slowest response times recorded.
	Median (med)	The midpoint (50% of requests are faster than this).
	Percentiles (p90, p95, p99)	The Gold Standard. P95 means 95% of users experienced this response time or faster. Reflects real user experience.
3. Stability	Error Rate (http_req_failed)	Percentage of failed requests
4. Workload	Max VUs (Virtual Users)	Peak number of concurrent users simulated during the test run

Why Percentiles Over Averages?

The Average	The Percentile Standard (P90 / P95 / P99)
How it works: • Adds all numbers and divides by total.	**How it works:** • Sorts all response times from fastest to slowest.
The Example (100 users): • 99 users get 100 ms (very fast) • 1 user waits 60,000 ms (stuck for 1 min) → **Average = ~700 ms**	**Same Example (Sorted 1 to 100):** • P95 = 100 ms → 95 users waited <= 100 ms • P99 = 100 ms → 99 users waited <= 100 ms → **Only 1 user had the problem.**
Why avoid it: • One bad user ruins the whole score. • It gives a false picture.	**Why use it:** • Filters out extreme noise. • Shows the real experience of normal users.

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Environment Setup



Demo Setup: Environment & Target API

Load Generator (Client)	Target API & Payload	Target System (SUT)
Tool: • K6	**Endpoint:** • POST /api/orders	**Architecture**: • Node.js (Express) + PostgreSQL 16 • Managed via Docker Compose
Host Execution: • Runs on Host OS (Laptop) • Sends requests to localhost:3000	**Payload:** {"item_name": "Demo Item","amount": 1}	**Resource Limits:** • App: 1.5 Cores, 1024 MB RAM • DB: 1.5 Cores, 2048 MB RAM **DB Pool:** • Max 10 connections

Test Scenarios



Load Test

Goal & Strategy	Workload Stages	Pass / Fail Criteria (Thresholds)
Objective: • Verify system behavior under expected normal peak traffic. **Test Type:** • Load Test. **Total Duration:** • 2 minutes 40 seconds. **Max Concurrency:** • 20 Virtual Users (VUs).	**1. Ramp-up (30s):** • 0 → 20 VUs. • Gradually warms up the server. **2. Steady State (2m):** • Maintain 20 VUs. • Measures stability and response time. **3. Ramp-down (10s):** • 20 → 0 VUs.	**1. Error Rate:** • http_req_failed: rate == 0% • Zero request failures allowed. **2. Response Time:** • http_req_duration: p(95) < 500ms • 95% of requests must complete under 500ms.

Stress Test

Goal & Strategy	Workload Stages	Pass / Fail Criteria (Thresholds)
Objective: • Push system beyond normal limits to find the breaking point. • Observe error behaviors and auto-recovery. **Total Duration:** • ~2 minutes 15 seconds. **Max Concurrency:** • 350 Virtual Users (VUs).	• **Step 1 (30s): Ramp to 50 VUs.** • **Step 2 (30s): Scale to 100 VUs.** • **Step 3 (30s): Push to 200 VUs.** • **Step 4 (30s): Peak at 350 VUs.** • **Step 5 (15s): Drop to 0 VUs**	**1. Error Rate:** • http_req_failed: rate < 15% • Allows small error rate before full collapse. **2. Response Time:** • http_req_duration: p(95) < 2000ms

Spike Test

Goal & Strategy	Workload Stages	Pass / Fail Criteria (Thresholds)
Objective: • Verify system resilience under sudden, extreme bursts of traffic. • Check if the server automatically recovers once load drops. **Total Duration:** • 1 minute 30 seconds. **Pattern:** • Normal → Sudden Spike → Cool-down.	**1. Baseline (15s): 10 VUs.** **2. Spike Surge (10s): Jump 10 → 200 VUs.** **3. Hold Peak (30s): Stay at 200 VUs.** **4. Cool-down (10s): Drop 200 → 10 VUs.** **5. Recovery Observation (20s):** **Maintain 10 VUs to check recovery.** **6. Drain (5s): 10 → 0 VUs.**	**1. Error Rate:** • http_req_failed: rate < 10% • Pool may drop requests during spike, but must not crash. **2. Response Time:** • http_req_duration: p(95) < 1500ms • 95% of requests must complete under 1500ms.


THANK YOU!